



### 3 Complex Analysis 2024

#### Tutorial 2

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## Limits

**Problem 6** If the limit exists, evaluate it. If the limit does not exist, prove this.

$$(a) \lim_{z \rightarrow 1} \frac{\bar{z} - 1}{z^2 - 1}, \quad (b) \lim_{z \rightarrow i} \frac{z - i}{z^2 + 1}.$$

## Differentiability, holomorphic functions

**Problem 7** Show from the definition of the derivative that the polynomial

$$f(z) = z^2 + 4iz - 4$$

is differentiable at  $z = i$ , and calculate the derivative  $f'(i)$ .

**Problem 8** We know that the function  $f(z) = z^2 + 4iz - 4$  is differentiable at any point because it is a polynomial. Show that it satisfies the Cauchy-Riemann equations at any point.

**Problem 9** At which points  $z \in \mathbb{C}$  is the function  $f(z) = z|z|$  differentiable? Where is this function holomorphic?

**Problem 10** Where is the function  $f(z) = \frac{1}{z^2 - 1}$  holomorphic?